

Doc 3: Vectors and Matrices

Module 0: Warm-up | Theory recap (read first, about 10 minutes)

1. Vectors: an ordered list of numbers

A vector is a list of numbers in a fixed order. The number of entries is its **length** (or dimension). In ML, one data point is a vector. A house with size 2, bedrooms 3 and age 1 becomes the vector (2, 3, 1), written as a column:

2
3
1

Length 3. Entry 1 is size, entry 2 is bedrooms, entry 3 is age.

Order matters: (2, 3, 1) is a different vector from (3, 2, 1).

Add two vectors	add entry by entry (same length only)	$(1, 2) + (3, 5) = (4, 7)$
Subtract	subtract entry by entry	$(4, 7) - (1, 2) = (3, 5)$
Scale by a number c	multiply every entry by c	$3 \times (1, 2) = (3, 6)$

2. Matrices: a grid of numbers

A matrix has **rows** (across) and **columns** (down). A matrix with m rows and n columns has **shape $m \times n$** . The entry A_{ij} sits in row i , column j : remember **Row first, then Column**.

a_{11}	a_{12}	a_{13}
a_{21}	a_{22}	a_{23}

This matrix has 2 rows and 3 columns, so its shape is **2×3** .

A dataset of 3 houses with 3 features each is a 3×3 matrix: each **row** is one house, each **column** is one feature.

Add / subtract	same shape only; entry by entry	shapes 2×3 and 2×3 : OK. 2×3 and 3×2 : not allowed
Scale by c	multiply every entry by c	$2 \times [1\ 2; 3\ 4] = [2\ 4; 6\ 8]$
Transpose A^T	rows become columns: $(A^T)_{ij} = A_{ji}$	a 2×3 matrix becomes 3×2

3. Where machine learning uses them

- **Dataset:** a matrix with one row per example and one column per feature.
- **Image:** a matrix of pixel brightness values (a colour image is three of them).
- **Neural network layer:** its weights form a matrix (Doc 5: matrix multiplication).
- **Word embeddings:** each word is a vector; a sentence is a matrix of these vectors.

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Worked examples: every cell filled in. Read them, then cover the answers and redo them.

Example 1: Shape and position

2	5	1
0	3	4

Shape: **2 rows $\times$ 3 columns = 2×3**

A_{12} = row 1, column 2 = **5**

A_{23} = row 2, column 3 = **4**

Row 2 is the vector (0, 3, 4). Column 1 is (2, 0).

Example 2: Vector arithmetic

$$\begin{array}{|c|} \hline 1 \\ \hline 2 \\ \hline 3 \\ \hline \end{array} + \begin{array}{|c|} \hline 4 \\ \hline 0 \\ \hline -1 \\ \hline \end{array} = \begin{array}{|c|} \hline 5 \\ \hline 2 \\ \hline 2 \\ \hline \end{array} \quad 2 \times \begin{array}{|c|} \hline 1 \\ \hline 2 \\ \hline 3 \\ \hline \end{array} = \begin{array}{|c|} \hline 2 \\ \hline 4 \\ \hline 6 \\ \hline \end{array}$$

a b a + b a 2a

Add or scale one position at a time: $1+4 = 5$, $2+0 = 2$, $3+(-1) = 2$. Then $2 \times 1 = 2$, $2 \times 2 = 4$, $2 \times 3 = 6$.

Example 3: Matrix arithmetic

$$\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array} + \begin{array}{|c|c|} \hline 5 & 0 \\ \hline -1 & 2 \\ \hline \end{array} = \begin{array}{|c|c|} \hline 6 & 2 \\ \hline 2 & 6 \\ \hline \end{array} \quad 3 \times \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array} = \begin{array}{|c|c|} \hline 3 & 6 \\ \hline 9 & 12 \\ \hline \end{array}$$

A B A + B A 3A

Same shape, same position: $1+5 = 6$, $2+0 = 2$, $3+(-1) = 2$, $4+2 = 6$. Scaling multiplies all four entries by 3.

Example 4: Transpose

$$\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & 5 & 6 \\ \hline \end{array} \quad \text{transpose} \quad \begin{array}{|c|c|} \hline 1 & 4 \\ \hline 2 & 5 \\ \hline 3 & 6 \\ \hline \end{array}$$

A (2×3) A^T (3×2)

Row 1 of A (1, 2, 3) becomes column 1 of A^T . Shape 2×3 becomes 3×2 . Transposing twice returns the original matrix.

Example 5: Is the operation allowed?

Operation	Allowed?	Result shape / reason
P (3×2) + Q (3×2)	Yes	3×2 : same shape
P (2×2) + Q (2×3)	No	shapes differ
Transpose of R (1×4)	Yes	4×1

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Exercises: use a pencil. Yellow cells are yours to fill in. Name: _____ Date: _____

A. Shape and position

M

4	0	7	2
1	6	3	9
5	8	2	0

Shape of M	
M_{23}	
M_{31}	
Row 2 of M	
Column 4 of M	

B. Vector arithmetic

i	1	2	3	4
a_i	2	-1	3	0
b_i	1	4	-2	5
$a + b$				
$a - b$				
$3a$				
$2a - b$				

C. Matrix arithmetic

$$\begin{array}{|c|c|c|} \hline 2 & -1 & 0 \\ \hline 3 & 4 & 1 \\ \hline \end{array} + \begin{array}{|c|c|c|} \hline 1 & 1 & 5 \\ \hline -2 & 0 & 3 \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array}$$

A B A + B

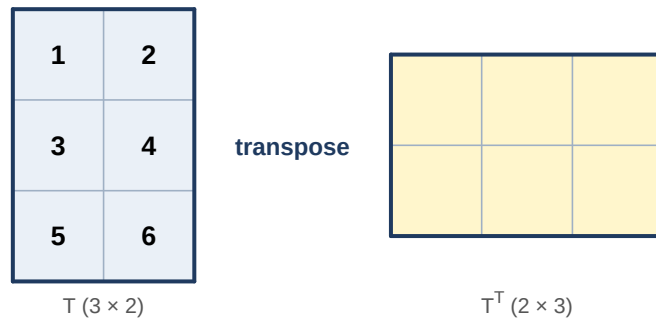
$$2 \times \begin{array}{|c|c|c|} \hline 2 & -1 & 0 \\ \hline 3 & 4 & 1 \\ \hline \end{array} - \begin{array}{|c|c|c|} \hline 1 & 1 & 5 \\ \hline -2 & 0 & 3 \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array}$$

A B 2A - B

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Exercises, continued

D. Transpose



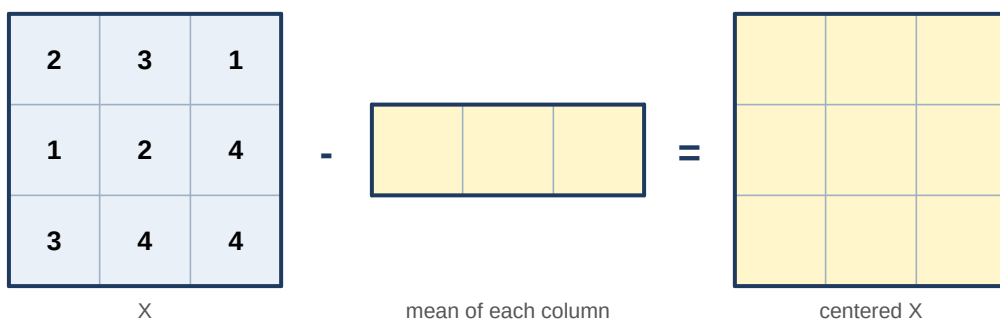
A matrix A has shape 5×2 . The shape of A^T is _____. $(A^T)^T =$ _____.

E. Is the operation allowed? If yes, give the result shape

Operation	Allowed? (Yes / No)	Result shape
$A \ (2 \times 3) + B \ (2 \times 3)$		
$A \ (2 \times 3) + B \ (3 \times 2)$		
$5 \times A$, where A is 4×2		
Transpose of $A \ (4 \times 2)$		
$a \ (3 \times 1) + b \ (1 \times 3)$		

F. Challenge: centering a dataset

Rows are 3 houses; columns are size, bedrooms, age. Step 1: find the **mean of each column** (add the 3 numbers, divide by 3) and write it in the yellow row. Step 2: subtract that row from **every** row of X .



Check: each column of centered X adds up to _____. (This centering is the first step of PCA in Doc 9.)

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Answer key: check your work only after finishing

A. Shape and position

Shape of M	3×4
M_{23}	3
M_{31}	5
Row 2 of M	(1, 6, 3, 9)
Column 4 of M	(2, 9, 0)

B. Vector arithmetic

i	1	2	3	4
a + b	3	3	1	5
a - b	1	-5	5	-5
3a	6	-3	9	0
2a - b	3	-6	8	-5

C. Matrix arithmetic

3	0	5
1	4	4

A + B

3	-3	-5
8	8	-1

2A - B

D and E

1	3	5
2	4	6

$T^T (2 \times 3)$

A^T has shape 2×5 . $(A^T)^T = A$.

Operation	Allowed?	Result shape
A (2×3) + B (2×3)	Yes	2×3
A (2×3) + B (3×2)	No	none
$5 \times A$, where A is 4×2	Yes	4×2
Transpose of A (4×2)	Yes	2×4
a (3×1) + b (1×3)	No	none

F. Centering

2	3	3
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column means

0	0	-2
-1	-1	1
1	1	1

centered X

Each column of centered X adds up to 0: $(0 - 1 + 1)$, $(0 - 1 + 1)$, $(-2 + 1 + 1)$.

Key takeaways: vectors are ordered lists; matrices are $m \times n$ grids (row first, then column); add and subtract need the same shape; transpose swaps rows and columns.

Next: Doc 4 - Vectors, dot product, norms.